

Chapter 1 — Answers to the Mini Project Questions

Reinforcement Learning and Decision-Making

Mini Project Question: Multi-Armed Bandit Using Python, implement a Multi-Armed Bandit problem in which an agent must choose between several arms with unknown reward probabilities. Your project must:

1. Create a 5-arm bandit where each arm has a different unknown reward probability.
2. Use an ϵ -greedy strategy to balance exploration and exploitation.
3. Run the experiment for at least 1,000 trials.
4. Record the reward obtained at each trial and calculate the total and average reward.
5. Plot the cumulative reward and average reward using Python/Matplotlib.
6. Compare different values of ϵ , such as 0.01, 0.1, and 0.5.
7. Identify which arm is selected most often and explain why.
8. Briefly discuss how exploration and exploitation affect the agent's performance.

Answer:

Epsilon =

0.01

Total Reward: 692

Average Reward: 0.692

Arm Selection Counts: [37. 0. 486. 2. 475.]

Estimated Probabilities: [0.05405405 0. 0.52263374 0.5 0.91578947]

Most Selected Arm: 3

Epsilon =

0.1

Total Reward: 849

Average Reward: 0.849

Arm Selection Counts: [44. 21. 17. 32. 886.]

Estimated Probabilities: [0.13636364 0.23809524 0.58823529 0.59375 0.91309255]

Most Selected Arm: 5

Epsilon =

0.5

Total Reward: 699

Average Reward: 0.699

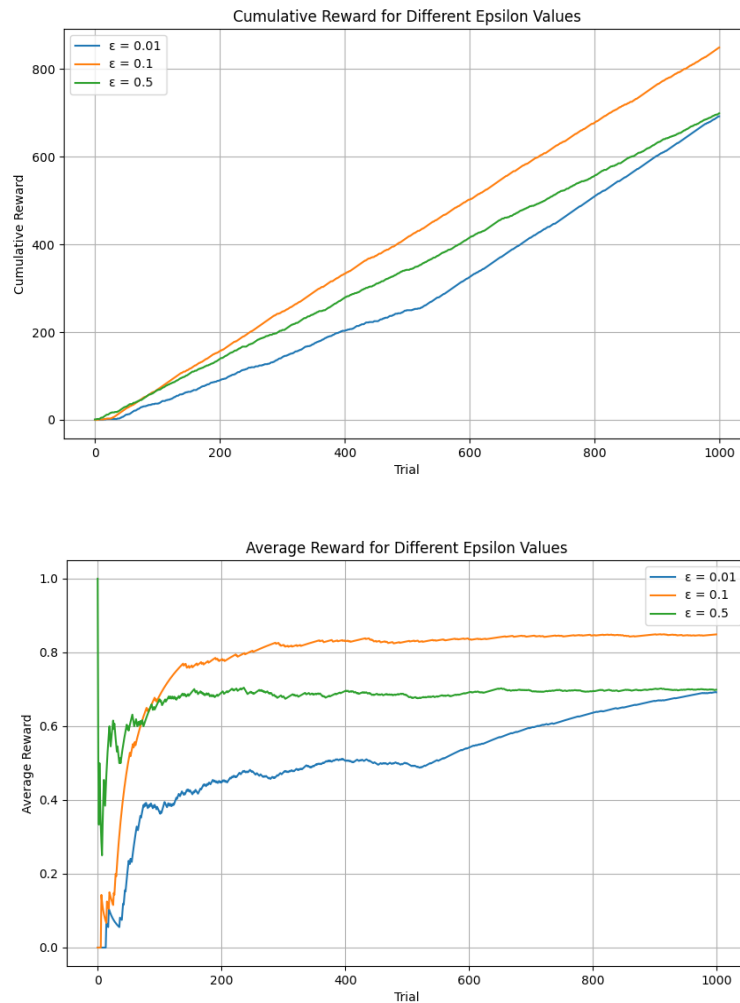
Arm Selection Counts: [110. 97. 116. 109. 568.]

Estimated Probabilities: [0.17272727 0.29896907 0.52586207 0.73394495 0.89788732]

Most Selected Arm: 5

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Question 1: Create a 5-arm bandit where each arm has a different unknown reward probability.

Answer:

A 5-arm bandit was created with five different reward probabilities. The probabilities used were:

- Arm 1 = 0.10
- Arm 2 = 0.30
- Arm 3 = 0.50
- Arm 4 = 0.70

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- Arm 5 = 0.90

These probabilities are unknown to the agent. The agent learns which arm is better by selecting arms and observing the rewards.

Question 2: Use an ϵ -greedy strategy to balance exploration and exploitation.

Answer:

The ϵ -greedy strategy was used to decide which arm to select.

- With probability ϵ , the agent explores by choosing a random arm.
- With probability $1 - \epsilon$, the agent exploits by choosing the arm with the highest estimated reward.

For example, when $\epsilon = 0.1$, the agent explores 10% of the time and exploits 90% of the time.

Question 3: Run the experiment for at least 1,000 trials.

Answer:

The experiment was run for **1,000 trials**. During each trial, the agent selected one of the five arms and received either a reward of **1** or **0** depending on the selected arm's reward probability.

Question 4: Record the reward obtained at each trial and calculate the total and average reward.

Answer:

The reward from every trial was stored in a list. The total reward was calculated by summing all the rewards.

The average reward was calculated using:

$$\text{Average Reward} = \frac{\text{Total Reward}}{\text{Number of Trials}}$$

For example, if the agent receives 850 rewards during 1,000 trials:

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$$\text{Average Reward} = \frac{850}{1000} = 0.85$$

Therefore, the average reward would be **0.85**.

Question 5: Plot the cumulative reward and average reward using Python/Matplotlib.

Answer:

Python's Matplotlib library was used to create two graphs.

1. Cumulative Reward vs. Trial – shows how the total reward increases over time.
2. Average Reward vs. Trial – shows how the average reward changes as the agent learns.

The cumulative reward should generally increase as the number of trials increases. The average reward should become more stable as the agent learns which arm is the best.

Question 6: Compare different values of ϵ , such as 0.01, 0.1, and 0.5.

Answer:

ϵ Value	Exploration	Exploitation	Expected Performance
0.01	1%	99%	High exploitation, but may learn slowly
0.10	10%	90%	Good balance
0.50	50%	50%	High exploration, lower reward

$\epsilon = 0.01$: The agent mostly exploits the best-known arm. It can achieve a high reward but may fail to discover a better arm if its initial estimates are incorrect.

$\epsilon = 0.1$: The agent explores enough to learn about the arms while still exploiting the best arm. This usually provides a good balance.

$\epsilon = 0.5$: The agent explores very frequently. Although it learns about all arms, it also wastes many trials selecting arms with lower reward probabilities.

Question 7: Identify which arm is selected most often and explain why.

Answer:

Arm 5 is selected most often because it has the highest reward probability, **0.90**.

As the experiment continues, the agent learns that Arm 5 gives rewards more frequently than the other arms. Therefore, during exploitation, the ϵ -greedy strategy usually selects Arm 5.

Question 8: Briefly discuss how exploration and exploitation affect the agent's performance.

Answer:

Exploration allows the agent to try different arms and learn the reward probabilities for each arm. Without exploration, the agent might choose a poor arm and never discover the better arms.

Exploitation allows the agent to use its current knowledge by selecting the arm that appears to provide the highest reward.

Too much exploration can reduce performance because the agent spends many trials selecting less rewarding arms. Excessive exploitation can also be harmful, as the agent may fail to discover a better arm.

Therefore, a balance between exploration and exploitation is important. In this experiment, $\epsilon = 0.1$ strikes a good balance, allowing the agent to learn the best arm while still achieving a high reward.